ELEC5803 HLS Exam Study
Step-by-Step Solution to Question 4

Course-slide basis

This solution follows the **sensitive path algorithm**, **fault coverage**, and **test coverage** definitions from `Slides/ELEC5803_Slides10_Testability and Dependability_AA2.pdf`.

1. Interpreting the circuit

From the gate symbols in the exam figure, the labeled nodes are interpreted as

$$\begin{aligned} D &= A \cdot B, \\ E &= B + C, \\ F &= B \cdot C, \\ H &= E \cdot F, \\ G &= D + H. \end{aligned}$$

So the circuit can be redrawn as:

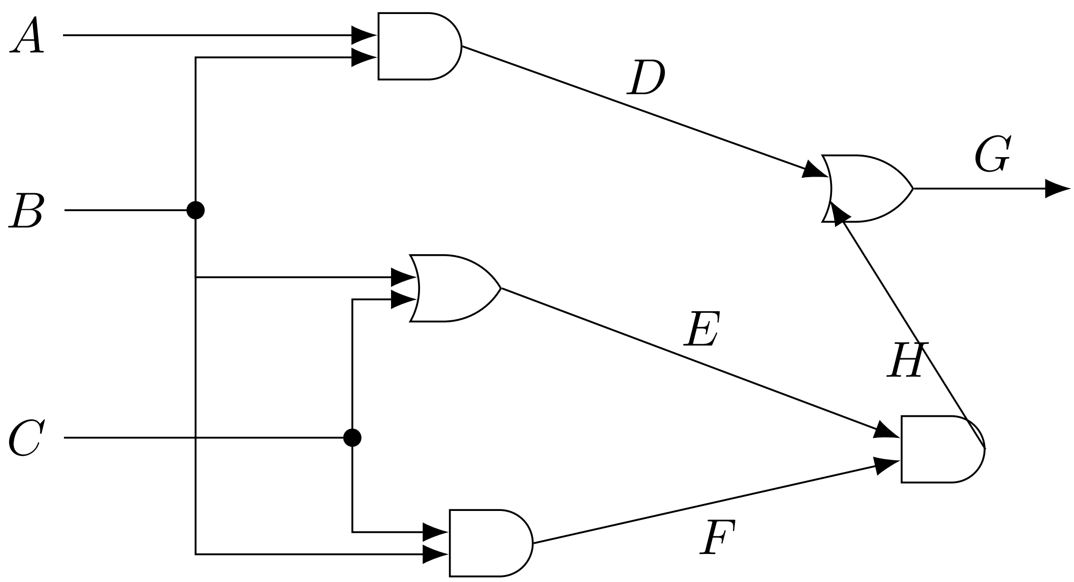

Figure 1: Native-LaTeX redraw of the Q4 circuit.

2. Stuck-at fault list

The question asks about points (A, B, C, D, E, F, G, H). Under the single stuck-at fault model, each point may be stuck at 0 or stuck at 1.

Therefore the full fault list has

$$8 \times 2 = 16$$

faults:

$$\{A/0, A/1, B/0, B/1, C/0, C/1, D/0, D/1, E/0, E/1, F/0, F/1, G/0, G/1, H/0, H/1\}.$$

3. Part (a): minimum test pattern set using the sensitive path algorithm

3.1 Sensitive path method from the slides

The slides summarize the method as:

1. select a fault,
2. force the faulty node to the opposite logic value in the fault-free circuit,
3. sensitize a path from that node to an observable output,
4. check consistency,
5. use one test to cover as many faults as possible.

3.2 Derive representative tests

Test 1: detect $A/0$ To detect $A/0$, the fault-free value at A must be 1, so set

$$A = 1.$$

Since A is only observable through

$$A \to D \to G,$$

we must make

$$B = 1$$

so that $D = A \cdot B$ depends on A.

Also, the other OR input must be 0 so that D is visible at G:

$$H = 0.$$

One simple choice is

$$C = 0 \Rightarrow F = B \cdot C = 0 \Rightarrow H = E \cdot F = 0.$$

Hence a valid test is

$$\boxed{110/10}$$

where the fault-free output is $(G, H) = (1, 0)$.

Test 2: detect $A/1$ To detect $A/1$, force the fault-free value

$$A = 0.$$

Again choose

$$B = 1$$

to let D depend on A, and choose

$$C = 0$$

to force $H = 0$ and sensitize the path to G.

So another valid test is

$$\boxed{010/00}.$$

Test 3: detect $E/0$ To detect $E/0$, the fault-free value at E must be 1:

$$E = B + C = 1.$$

To observe E through

$$E \to H \to (H, G),$$

the other input of the AND gate must be 1:

$$F = 1.$$

Since

$$F = B \cdot C,$$

this requires

$$B = 1, \quad C = 1.$$

Using

$$A = 0$$

avoids unnecessary masking through D.

Thus a valid test is

$$\boxed{011/11}.$$

Test 4: detect $B/1$ To detect $B/1$, the fault-free value at B must be 0:

$$B = 0.$$

Choose

$$A = 0, \quad C = 1.$$

Then in the fault-free circuit:

$$D = 0, \quad E = 1, \quad F = 0, \quad H = 0, \quad G = 0.$$

If B is stuck at 1:

$$E = 1, \quad F = 1, \quad H = 1, \quad G = 1.$$

So the fault is observable at both outputs.

Hence a valid test is

$$\boxed{001/00}.$$

3.3 Minimum test set

Collecting the four tests above:

$$\boxed{T = \{001/00,\ 010/00,\ 011/11,\ 110/10\}.}$$

This set detects all detectable faults of the circuit.

3.4 Faults covered by each test

Test vector	Faults detected
001/00	$B/1,\ D/1,\ F/1,\ G/1,\ H/1$
010/00	$A/1,\ C/1,\ D/1,\ F/1,\ G/1,\ H/1$
011/11	$B/0,\ C/0,\ E/0,\ F/0,\ G/0,\ H/0$
110/10	$A/0,\ B/0,\ C/1,\ D/0,\ F/1,\ G/0,\ H/1$

3.5 Untestable fault

The only undetectable fault is

$$\boxed{E/1}.$$

Reason: To detect $E/1$, the fault-free circuit must have

$$E = 0.$$

But to propagate E through the AND gate to H, we must also have

$$F = 1.$$

Since

$$F = B \cdot C,$$

$F = 1$ implies

$$B = 1, \quad C = 1.$$

However then

$$E = B + C = 1,$$

which contradicts the requirement that the fault-free value of E be 0.
Therefore no sensitive path test exists for $E/1$.

4. Part (b): fault coverage and test coverage

From the slides:

$$FC = \frac{N_{\text{det}}}{N} \times 100\%, \qquad TC = \frac{N_{\text{det}}}{\hat{N}} \times 100\%$$

where:

- N = total number of faults,
- N_{det} = number of detected faults,
- $\hat{N}$ = number of detectable faults.

4.1 Count the faults

$$N = 16.$$

Since only $E/1$ is undetectable,

$$\hat{N} = 15.$$

Our test set detects all detectable faults, so

$$N_{\text{det}} = 15.$$

4.2 Fault coverage

$$FC = \frac{15}{16} \times 100\% = 93.75\%.$$

4.3 Test coverage

$$TC = \frac{15}{15} \times 100\% = 100\%.$$

4.4 Final answer for part (b)

$$\boxed{FC = 93.75\%, \qquad TC = 100\%.}$$

5. Part (c): observability and testability of each point

Point	Observability / testability	Discussion
A	Good controllability, medium observability	As a primary input, A is easy to set to 0 or 1. It is observable only through D and then G, so the path must be sensitized by forcing $B = 1$ and $H = 0$. Both $A/0$ and $A/1$ are detectable.
B	Good controllability, good observability	B fans out to D, E, and F, so changes at B can affect both H and G. This gives several detection opportunities, and both stuck-at faults are detectable.
C	Good controllability, medium observability	C affects E and F, then reaches outputs through H and G. It is less directly observable than B, but both $C/0$ and $C/1$ are still detectable.
D	Good observability when $H = 0$	D reaches the output only through the final OR gate, so to observe D we must force the other OR input to 0. Hence D is reasonably testable, but only after sensitization.
E	Poor observability, poorest testability	E is observable only through $H = E \cdot F$, so we need $F = 1$ to make E visible. This allows testing $E/0$, but makes $E/1$ impossible to test. Thus E is the weakest point in the circuit.
F	Good observability when $E = 1$	F can be observed at H and then at G, provided the other AND input is set to 1. Both stuck-at faults are detectable.
G	Highest observability	G is a primary output, so any fault effect reaching G is directly visible. Both $G/0$ and $G/1$ are easy to test.
H	Very high observability	H is directly labeled as an observable node and also feeds G. This makes it one of the easiest internal nodes to observe and test.

6. Final exam-style answers

(a)

Using the sensitive path algorithm, a minimum test set is

$$\boxed{T = \{001/00,\ 010/00,\ 011/11,\ 110/10\}.}$$

(b)

The set detects 15 of the 16 stuck-at faults. Since only $E/1$ is undetectable:

$$\boxed{FC = 93.75\%, \qquad TC = 100\%.}$$

(c)

Among all points, E has the poorest observability and testability because one of its stuck-at faults ($E/1$) is untestable. Points G and H have the best observability because they are directly observable outputs / output-near nodes.